

RAG Practice Knowledge Base

A sample PDF for practicing LangChain PDF loading, document inspection, chunking, embeddings, vector stores, retrieval, and question answering.

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1. What is Retrieval-Augmented Generation?

Retrieval-Augmented Generation, commonly called RAG, is an approach that combines information retrieval with a large language model. Instead of relying only on information learned during model training, a RAG application searches an external knowledge source for relevant information and supplies that information to the language model as context.

A typical RAG system has two broad phases. During indexing, documents are loaded, split into smaller chunks, converted into numerical embeddings, and stored in a searchable database. During retrieval, a user's question is converted into a representation that can be compared with stored chunks. Relevant chunks are returned and included in the prompt sent to the language model.

2. Document Loading

Document loading is the ingestion step. A loader reads information from a source and converts it into a standard document representation. Sources can include text files, PDFs, web pages, CSV files, Word documents, or entire directories. In LangChain, a Document normally contains `page_content` and `metadata`. The `page_content` stores the text, while `metadata` can identify properties such as the source filename or page number.

Field	Example	Purpose
page_content	Machine learning is...	Stores extracted text
source	training_manual.pdf	Identifies the original file
page	2	Identifies a page or section

3. Text Splitting and Chunking

Large documents are usually divided into smaller pieces called chunks. Chunking is important because language models and embedding models operate within practical context limits, and retrieval works better when the stored units are focused. A chunk size determines approximately how much content belongs to one chunk. Chunk overlap allows neighboring chunks to share some text so that information near a boundary is less likely to lose context.

For example, imagine a document containing 5,000 characters. A developer might experiment with chunks of 1,000 characters and an overlap of 150 characters. These values are not universal rules. The best strategy depends on the structure of the source material, the embedding model, and the questions users are expected to ask.

4. Embeddings

An embedding is a numerical representation of content. Texts with related meanings tend to have representations that are closer to each other according to a similarity measure. In a RAG application, embeddings make it possible to search by semantic meaning rather than requiring exact keyword matches.

Suppose the knowledge base contains the sentence 'Python is commonly used for data science.' A user could ask 'Which programming language is popular for data analysis?' Even though the wording is different, semantic retrieval may identify the Python passage as relevant.

5. Vector Stores and Retrieval

A vector store keeps embeddings together with their associated text and metadata. When a question arrives, the system searches for vectors that are similar to the question's embedding. The retrieved text can then be passed to the language model. Common experiments include changing the number of retrieved chunks, filtering by metadata, and comparing similarity-search strategies.

6. Fictional Company Information

Northstar Analytics is a fictional software company founded in 2021. Its headquarters are in Austin, Texas. The company develops data-analysis tools for small logistics businesses. Its flagship product is called RouteLens.

RouteLens helps logistics teams analyze delivery performance, estimate route delays, and create weekly operational reports. The standard subscription costs \$79 per month for up to five users. The professional subscription costs \$149 per month for up to twenty users. Customers can cancel a monthly subscription at any time before the next billing date.

Northstar Analytics provides customer support Monday through Friday from 9:00 AM to 6:00 PM Central Time. The fictional support email used in this training document is support@example.invalid. Critical enterprise incidents are assigned a target initial response time of two hours.

7. Employee Handbook Excerpt

Employees receive twenty days of paid annual leave each calendar year. Leave requests longer than five consecutive working days should normally be submitted at least fourteen days in advance. Employees may work remotely up to three days per week when their role and team schedule permit it.

The company requires multi-factor authentication for access to production systems. Passwords must not be shared between employees. Suspected credential exposure should be reported to the security team as soon as it is discovered.

Practice Questions

1. What does RAG stand for?
2. What are the two main pieces commonly stored in a LangChain Document?
3. Why is chunk overlap useful?
4. What is the name of Northstar Analytics' flagship product?
5. How much does the professional subscription cost?
6. How many paid annual leave days do employees receive?
7. What security control is required for production-system access?
8. On which page does the fictional company information appear?